Cognitive Physics: A Measurement Program
Trait–State Covariance Geometry in Jointly Observed Systems

Joel Peña Muñoz Jr.

October 3, 2026

Abstract

Complex systems can be studied through the geometry of persistence and the movement of state. This paper develops Cognitive Physics as a falsifiable measurement program for examining those two aspects of variation within genuinely jointly observed systems. A working trait–state decomposition motivates separate covariance estimates for unit means and within-unit deviations. The central hypothesis predicts that between-unit covariance occupies fewer effective dimensions than within-unit covariance under common, prespecified feature scaling. The first prospective test used eyes-open resting EEG from the PhysioNet EEG Motor Movement/Imagery Dataset, with participants as units and disjoint recording epochs as repeated observations in a ten-feature representation. After quality control, the analysis retained 78 participants and 1,322 epochs. The author-reported run yielded participation-ratio dimensionalities of 3.54 between participants and 6.01 within participants, giving $\widehat{\Delta}_d = 2.47$ with a bootstrap 95% interval of $[1.92, 3.01]$ and supporting the prespecified ordering. Leave-one-participant-out estimates remained stable at approximately 2.47 ± 0.04. A participant-label shuffle reduced the mean difference to approximately 0.83, while random one-dimensional projections produced a difference near zero, indicating that participant identity and the multivariate feature design both contribute to the measured separation. These findings establish a result within the declared dataset, quality-control procedure, and feature space. Raw-band and feature ablations, quality-control sensitivity analyses, independent-session reliability, held-out prediction, and controlled intervention studies will evaluate its robustness, persistence, utility, and selective response. Building on established multilevel covariance decomposition and participation-ratio dimensionality, the methodological contribution is to formulate their within-versus-between difference as a hierarchy-preserving, uncertainty-quantified directional test. The broader contribution is a traceable path from joint measurements to covariance geometry and independently tested interpretations.

Keywords: Cognitive Physics; trait–state decomposition; covariance geometry; effective dimensionality; joint measurements; EEG.

1 Introduction

A complex system offers two views of its variation: the structure that persists across observations and the fluctuations that unfold within them. Cognitive Physics is developed here as a measurement program that brings these views into a common mathematical frame. Its guiding question asks whether genuinely joint measurements reveal a reproducible difference between the covariance geometry of observational units and the geometry of their movement over time. Persistence and fluctuation thereby become questions of sampling, dimensionality, reliability, and response.

The investigation follows a three-stage sequence. Jointly sampled observations establish the empirical ground; between-unit and within-unit covariance describe its geometry; prediction and intervention studies examine the meaning and utility of the resulting patterns. Relatively persistent between-unit structure provides a candidate trait component, while time-varying within-unit structure provides a candidate state component. Repeated measurements, documented uncertainty, sensitivity analyses, and independent replication determine how far these interpretations can travel.

The observational unit forms the foundation of the comparison. A participant contributes repeated recording epochs, while a model training run may contribute repeated checkpoints. Each row of a feature matrix preserves that identity through its sampling window, condition, and alignment procedure. Joint covariance rests on this shared observational ground. Each system is measured first in its own setting; comparisons across systems then address clearly defined properties of the resulting geometries.

The central hypothesis predicts that relatively persistent between-unit variation occupies fewer effective dimensions than within-unit fluctuations under a common feature representation and scaling rule. The first prospective test evaluated this ordering in eyes-open resting EEG from the PhysioNet EEG Motor Movement/Imagery Dataset. The author-reported analysis retained 78 participants and 1,322 epochs and yielded $d_{\mathrm{PR}} = 3.54$ between participants and $d_{\mathrm{PR}} = 6.01$ within participants, with $\widehat{\Delta}_d = 2.47$ and a bootstrap 95% interval of $[1.92, 3.01]$. Leave-one-participant-out estimates remained stable, while participant-label shuffling reduced the contrast and one-dimensional projections collapsed it toward zero. These findings support the prespecified ordering within the declared ten-feature representation and reveal where the next questions begin. Archive linkage, feature ablations, quality-control sensitivity analyses, independent-session reliability, prediction, and intervention will test the result's robustness and scientific reach.

2 Relation to Existing Work and Methodological Contribution

Three ingredients of the present framework have strong precedents. First, multilevel principal component methods decompose repeated observations into between-subject and within-subject covariance components. Multilevel functional principal component analysis formalized this decomposition for longitudinal functional data and applied it to sleep EEG [2]. Later multilevel EEG methods extended the same general separation to region-referenced functional signals and multivariate covariance structures [3]. The covariance decomposition in Equations (5) and (6) therefore belongs to an established statistical lineage.

The earlier sleep-EEG result also approached the empirical pattern studied here. At the visit-within-subject level, four components were required to explain 50% of variation and fourteen were required to explain 90%, which the authors interpreted as substantial within-subject heterogeneity [2]. This threshold-based summary already suggested that within-subject variation may spread across more directions than between-subject variation.

Second, psychometric trait–state research has long separated within-person state structure from between-person trait structure. Hamaker, Schuurman, and Zijlmans explicitly examined whether within-person state dimensions share the same factor structure as between-person trait dimensions [4]. The trait–state language used here therefore names an interpretive question with a substantial prior literature.

Third, participation ratio is an established continuous dimensionality measure. It is widely used to summarize neural covariance spectra [5, 6]. In contrast to counting the number of principal components needed to cross a selected variance threshold, participation ratio uses the full eigenspectrum:

$$d_{\mathrm{PR}}(\Sigma) = \frac{[\mathrm{tr}(\Sigma)]^2}{\mathrm{tr}(\Sigma^2)}. \tag{1}$$

Its value remains tied to representation, scaling, sampling, noise, and finite-sample estimation.

The candidate methodological contribution lies in joining these established ingredients into one primary falsifiable statistic:

$$\boxed{\Delta d_{\mathrm{PR}} = d_{\mathrm{PR}}(\Sigma_{\mathrm{within}}) - d_{\mathrm{PR}}(\Sigma_{\mathrm{between}})}. \tag{2}$$

The comparison is performed under common feature scaling and equal unit weighting, with uncertainty estimated by resampling units together with their repeated observations. A positive interval supports the proposed ordering, a negative interval supports a reversed ordering, and an interval crossing zero leaves the comparison unresolved. This construction turns two covariance spectra into a directly testable repeated-measures geometry claim.

Based on the neighboring literature reviewed here, we have yet to identify prior work that centers the within-versus-between participation-ratio difference as its primary uncertainty-quantified statistic. A defensible novelty statement is therefore: building on multilevel covariance decomposition and participation-ratio effective dimensionality, this study formulates their difference as an explicit directional hypothesis for repeated multivariate measurements. This statement concerns the construction and testing framework. Historical priority would benefit from a formal systematic review across multilevel PCA, functional data analysis, psychometrics, neural dimensionality, and effective-rank terminology.

3 Central Hypothesis

3.1 Observation model

Each joint observation places a system within a space of measured features. Let $\mathbf{X}_{s,t} \in \mathbb{R}^p$ denote the vector of p features collected for observational unit s at repeated observation t. Repeated observations trace the unit's passage through that space. A unit may be a participant or a model training run; its identity and sampling interval anchor the interpretation within each system. The working decomposition is

$$\mathbf{X}_{s,t} = \boldsymbol{\mu} + \mathbf{T}_s + \mathbf{S}_{s,t} + \boldsymbol{\varepsilon}_{s,t}, \tag{3}$$

Here $\boldsymbol{\mu}$ provides the population reference, $\mathbf{T}_s$ represents a candidate persistent displacement associated with the unit, $\mathbf{S}_{s,t}$ describes its time-varying state, and $\boldsymbol{\varepsilon}_{s,t}$ carries measurement error and residual variation. The decomposition brings persistence, movement, and measurement into a shared representation. Centering conventions place the population-average trait in $\boldsymbol{\mu}$ and the unit-specific average state over the reference sampling distribution in $\mathbf{T}_s$. These terms begin as operational components of the model. Identification assumptions, repeated-session measurements, and reliability assessments connect them to empirical interpretations and establish the timescales over which persistence can be evaluated.

3.2 Between-unit and within-unit geometry

The mean of each unit marks its average position over the sampled observations. Variation across these positions describes between-unit geometry; deviations around each position describe within-unit geometry. For N units with n_s observations each, define

$$\overline{\mathbf{X}}_s = \frac{1}{n_s}\sum_{t=1}^{n_s}\mathbf{X}_{s,t}, \qquad \overline{\mathbf{X}} = \frac{1}{N}\sum_{s=1}^{N}\overline{\mathbf{X}}_s. \tag{4}$$

With the observational hierarchy preserved, these two views of variation yield the empirical covariance geometries

$$\widehat{\Sigma}_{\text{between}} = \frac{1}{N-1}\sum_{s=1}^{N}(\overline{\mathbf{X}}_s - \overline{\mathbf{X}})(\overline{\mathbf{X}}_s - \overline{\mathbf{X}})^{\mathsf{T}}, \tag{5}$$

$$\widehat{\Sigma}_{\text{within}} = \frac{1}{N}\sum_{s=1}^{N}\left[\frac{1}{n_s-1}\sum_{t=1}^{n_s}(\mathbf{X}_{s,t} - \overline{\mathbf{X}}_s)(\mathbf{X}_{s,t} - \overline{\mathbf{X}}_s)^{\mathsf{T}}\right]. \tag{6}$$

These definitions require $N \geq 2$ and $n_s \geq 2$ and give each unit equal weight. A common set of feature coordinates and a prespecified scaling place both geometries on the same measurement ground. Their dimensionality comparison remains tied to that shared representation. The between-unit estimate includes finite-sample fluctuations in unit means, and the within-unit estimate includes state and residual variation. Both matrices therefore characterize observed covariance geometry. A measurement-error model, temporal sampling account, and independent reliability evidence provide the bridge from those matrices to latent trait–state interpretation.

3.3 Falsifiable dimensionality prediction

The covariance spectrum describes how variation gathers along measured directions. Its concentration can be summarized by the established participation-ratio effective dimensionality [5, 6]. For a nonzero positive semidefinite covariance matrix Σ with eigenvalues $\lambda_1, \ldots, \lambda_p$, define

$$d_{\mathrm{PR}}(\Sigma) = \frac{\left(\sum_{j=1}^{p} \lambda_j\right)^2}{\sum_{j=1}^{p} \lambda_j^2} = \frac{[\mathrm{tr}(\Sigma)]^2}{\mathrm{tr}(\Sigma^2)}. \tag{7}$$

A spectrum concentrated in a few directions yields a smaller effective dimensionality; variance distributed more evenly across directions yields a larger value. The covariance trace records total variance, and the participation ratio describes its directional distribution. The central hypothesis proposes a more concentrated between-unit geometry alongside a more distributed within-unit geometry. Evaluated separately within each prespecified system, the prediction is

$$\Delta_d \equiv d_{\mathrm{PR}}(\Sigma_{\mathrm{within}}) - d_{\mathrm{PR}}(\Sigma_{\mathrm{between}}) > 0. \tag{8}$$

Here $\Sigma_{\mathrm{between}}$ and Σ_{within} denote the population counterparts of the empirical geometries above. The hypothesis specifies an ordering and leaves both dimensionalities to measurement; each system retains the values produced by its feature representation, scaling, sampling, and covariance estimators. An uncertainty interval entirely above zero supports the proposed ordering, an interval entirely below zero supports the reversed ordering and challenges the hypothesis in that system, and an interval spanning zero leaves the comparison unresolved.

The first EEG test produced $\widehat{\Delta}_d = 2.47$, from within-participant and between-participant estimates of 6.01 and 3.54, respectively. Its participant-level bootstrap interval, $[1.92, 3.01]$, lies above zero and therefore supports the central ordering under the prespecified decision rule. This outcome constitutes one measured instance of the hypothesis in the declared ten-feature space. A future jointly sampled system may yield a similar, unresolved, or reversed ordering, with each result revising the empirical scope of the program. Dedicated studies of temporal persistence, predictive utility, and intervention response will establish how far the observed geometry can carry its trait–state interpretation.

4 Measurement Principles

4.1 Joint observations before joint covariance

Every feature vector carries an observational identity: a unit, a condition, and a window in time. These coordinates provide the empirical foundation of joint covariance. Each row of the EEG feature matrix joined ten measurements from one participant and one accepted two-second epoch. Participant identifiers preserved the hierarchy used to estimate within-participant covariance, between-participant covariance, participant-level uncertainty, and leave-one-participant-out sensitivity.

The completed run retained 78 participants and 1,322 epochs after quality control. Its exclusion record identified 31 participant IDs with too few accepted epochs for the declared comparison. A participant-label shuffle reduced $\widehat{\Delta}_d$ from 2.47 to a mean near 0.83, showing that the correspondence

between participants and their epochs contributes to the measured separation. Future joint matrices will carry the same visible path from source recording through acceptance decisions to numerical feature vectors.

4.2 Measure each system first; compare geometries second

Each system enters the study through its own measurements and timescales. Between-unit and within-unit geometries are estimated first within that setting, with feature definitions, units, scaling, sampling intervals, and retention rules reported alongside the spectra and effective dimensionalities. Comparisons across systems then address explicitly defined geometric summaries, each anchored to its original joint sample.

The EEG result illustrates why representation belongs inside the claim. Random one-dimensional projections produced a dimensionality difference near zero, while the ten-feature matrix yielded $\widehat{\Delta}_d = 2.47$. The observed ordering therefore describes multivariate structure expressed through the chosen measurement instrument. Comparisons of principal directions require justified correspondence between feature coordinates, and shared-mechanism interpretations require additional system-specific evidence. Similarities, differences, and reversed orderings each retain their place in the empirical account.

4.3 Prespecified methods and empirical outcomes

The protocol establishes the frame within which the measurements can speak. For the primary EEG run, observational units, the ten-feature representation, preprocessing, scaling, covariance estimators, exclusion criteria, participant-level bootstrap, and decision rule defined that frame. The report presents the estimated sign and uncertainty of Equation (8) together with the individual dimensionalities. Primary results and subsequent robustness analyses retain distinct labels, and any protocol deviations will enter the archival record.

Uncertainty analysis follows the structure of the observations. The primary bootstrap resampled participants together with their complete accepted epoch sequences, while leave-one-participant-out analysis examined influence within the retained cohort. Each permutation, projection, or surrogate analysis must identify the relationship it disrupts and the properties it preserves. Predictive validation will reserve entire units for evaluation, with data-dependent transformations fitted exclusively on training data. This separation preserves the distinction between observations used to construct a model and observations used to assess its reach.

4.4 Persistence, verification, and an auditable record

Persistence becomes an empirical property through repeated observation. The completed R01 analysis establishes a within-recording geometric result. Independent sessions or prespecified longitudinal intervals will connect between-unit covariance to measured reliability over time. Intervention-related state shifts and relative trait stability will receive their own comparison designs and prespecified

effect or equivalence criteria. Approximate invariance will be evaluated against a defined equivalence margin with sufficient precision to resolve that comparison.

Each reported result remains traceable to identifiable source data, feature-extraction code, analysis settings, and inclusion records. The current numerical summary retains author-reported status while the final JSON output, frozen implementation, exclusion ledger, and joint feature matrix are linked to the manuscript archive. Reproduction from those records will establish computational traceability; replication on separately sampled data will establish empirical reach. The record will preserve both the measured pattern and the conditions under which it appeared.

5 Reported Measurements, Primary Run, and Verification Plan

The measurement record brings together four strands of evidence: the completed primary EEG run, earlier author-reported within-system analyses, domain-specific benchmarks, and claims assigned to future validation. Each strand retains its observational setting and stage of verification. The completed EEG comparison now carries the primary evidential role for Equation (8); earlier numerical summaries remain part of the historical record and require reconstruction in their original analytical lineages. A traceable numerical record includes both the estimate and the meaning of its uncertainty. Reported $\pm$ values are transcribed as supplied, with their interpretation as standard deviations, standard errors, or other summaries to be established from the analysis records. Intervals labeled CI likewise require a documented confidence level and construction method.

5.1 Completed primary EEG record

The author-reported PhysioNet R01 run retained 78 participants and 1,322 accepted epochs in the ten-feature matrix. It produced

$$d_{\mathrm{PR}}(\widehat{\Sigma}_{\mathrm{between}}) = 3.54, \qquad d_{\mathrm{PR}}(\widehat{\Sigma}_{\mathrm{within}}) = 6.01, \qquad \widehat{\Delta}_d = 2.47, \tag{9}$$

with a participant-level bootstrap 95% interval of $[1.92, 3.01]$. Leave-one-participant-out analysis gave approximately 2.47 ± 0.04, participant-label shuffling reduced the mean contrast to approximately 0.83, and random one-dimensional projections yielded contrasts near zero. Section 8 presents the scientific interpretation of these results.

The manuscript archive currently includes the supplied numerical summary and a partial execution log. Full computational verification will link the final `RESULT_LATEST.json`, code revision, software environment, random seeds, input manifest, exclusion ledger, joint feature matrix, and generated uncertainty outputs. The primary values therefore retain author-reported status during archive completion. Once linked, a clean rerun from the frozen record will test exact numerical reproduction.

5.2 Joint matrices and shuffled comparators

Each joint matrix records a pattern of co-observation. In the earlier supplied summary, EEG features share a participant and recording epoch, cardiac features share a patient with paired Holter and echocardiographic measurements, and machine-learning features share a model. These legacy observational identities anchor three reported matrix summaries that remain separate from the completed R01 primary run:

System	Matrix	d_{PR}	PC1	PC2	Shuffled comparator
EEG legacy	1740×10	1.480	81.9%	3.5%	9.949 ± 0.011
Cardiac	120×7	1.638	77.6%	5.2%	6.663 ± 0.090
ML	80×5	1.331	86.4%	4.4%	4.767 ± 0.090

The reported EEG variance fraction for PC1 and PC2 together is 85.4%. The cardiac feature families combine Holter-derived measures with echocardiography. The ML summary identifies centered kernel alignment (CKA), judge correlation r_{judge}, and the scaling exponent N_{exp} among its five features. Reconstructing these matrices requires complete feature definitions and identification of the remaining ML columns. Each shuffled comparator will receive an explicit null interpretation through its permutation scheme, preserved marginal and temporal structure, and number of surrogate realizations.

Resampling provides a second view of each reported estimate. The supplied bootstrap summaries retain their own labels and intervals, alongside the original matrix estimates:

System	Bootstrap summary	Reported CI
EEG	1.480 ± 0.019	$[1.447, 1.516]$
Cardiac	1.632 ± 0.103	$[1.452, 1.822]$
ML	1.332 ± 0.060	$[1.233, 1.472]$

For the legacy EEG matrix, the reported leave-one-subject-out summary is 1.667 ± 0.015, with range 1.653–1.719. These values describe a different analytical lineage from the completed primary run's leave-one-participant-out contrast of approximately 2.47 ± 0.04. Matrix provenance, scaling, estimator definitions, and exclusion procedures will establish the source of each estimate and preserve their distinct meanings. Independent replication will extend sensitivity assessment to a separate sample.

5.3 Reported trait–state covariance comparisons

The repeated-observation hierarchy gives the measurement record three complementary views: full variation, fluctuations around unit means, and differences among those means. The supplied full, within-centered, and between-unit summaries below retain their identities as distinct reported analyses. Their relationship to the preceding matrix summaries will be established through matrix provenance, preprocessing, and estimator records.

	Full		Within-unit		Between-unit	
System	d_{PR}	PC1	d_{PR}	PC1	d_{PR}	PC1
EEG legacy	2.141	66.5%	9.727	13.7%	1.502	80.4%
Cardiac	1.271	88.4%	3.949	29.7%	1.078	96.3%
ML	1.230	89.9%	2.996	34.7%	1.018	99.1%

The cardiac design is described as 40 units observed at three times, and the ML design as 10 families with eight checkpoints each. The ML hierarchy requires an explicit definition of the repeated-observation unit as a family, model, or training run. The legacy EEG entry retains its participant-by-epoch hierarchy. Within the supplied record, all three pairs show the proposed ordering: variance across unit means is more concentrated than variance around those means. Their uncertainty and provenance remain to be reconstructed. The completed R01 test supplies the present primary EEG comparison, with 6.01 within participants, 3.54 between participants, and a paired bootstrap interval for their difference.

Numerical identity follows analytical lineage. The legacy EEG matrix estimate 1.480, legacy between-unit estimate 1.502, and completed R01 between-participant estimate 3.54 belong to separately reported analyses. The cardiac matrix estimate 1.638 and full repeated-measurement estimate 1.271 likewise describe distinct analyses, with analogous ML estimates of 1.331 and 1.230. Their matrix versions, feature transformations, and estimator outputs will provide the record needed to interpret each value in its proper setting.

5.4 Spectral summaries and entropy conventions

The eigenspectrum describes how the reported variance is apportioned among principal directions. For the joint-matrix analyses, the first four supplied normalized eigenvalues, $\tilde{\lambda}_j = \lambda_j / \sum_k \lambda_k$, are

$$\begin{aligned} \text{EEG:} &\quad (0.819,\ 0.035,\ 0.026,\ 0.024), \\ \text{Cardiac:} &\quad (0.776,\ 0.052,\ 0.048,\ 0.037), \\ \text{ML:} &\quad (0.864,\ 0.044,\ 0.034,\ 0.030). \end{aligned}$$

These four-mode summaries provide a partial view of each spectrum; reproducing the participation ratio requires the complete unrounded eigenvalues. The reported between-unit spectral-decay fits, $\lambda_j \sim j^{-\alpha}$, give $\alpha = 3.92$ for EEG, 3.61 for cardiac data, and 4.90 for ML. Each exponent retains its system-specific setting, with fitting range and uncertainty to be documented in the corresponding analysis record.

The supplied entropy and stability summaries describe additional aspects of concentration and fluctuation: trait entropy 0.065, state entropy 2.30, an EEG entropy score 0.66, and metastability 0.154 ± 0.057. Their interpretation follows the covariance or signal being measured and the estimator used. A common-scale comparison requires explicit entropy definitions, logarithm bases, and normalization rules. Eigenvalue entropy, permutation entropy, and temporal variability of synchrony will retain distinct measurement identities.

5.5 Longitudinal, cross-system, and intervention summaries

Longitudinal observations extend the question of persistence across visits. For the reported sample of 58 participants observed at three visits months apart, the supplied trait correlations are

$$\rho(t_0, t_1) = 0.969, \qquad \rho(t_0, t_2) = 0.919, \qquad \rho(t_1, t_2) = 0.876. \tag{10}$$

The reported state-entropy correlation is 0.018. Together, these summaries motivate an examination of the different temporal profiles of candidate trait and state measures. Their feature definitions, visit intervals, uncertainty, and adjustment procedures will establish the basis for evaluating repeatability over the sampled months.

The neuro-cardiac summary brings modalities into a shared patient-level setting. The paired EEG, ECG, and echocardiographic records cover 58 patients. The reported trait correlation is $\rho(\text{trait}_{\text{EEG}}, \text{trait}_{\text{cardiac}}) = 0.297$. Each modality retains its documented sampling window. Patient identifiers and temporal alignment will establish the correspondence across modalities, while cohort records will clarify the relationship to the longitudinal sample.

The intervention summary adds a reported perturbation to the observational record. For 30 participants measured before and after transcranial direct-current stimulation, it gives trait correlation 0.998 and mean state change 0.124 ± 0.212. Paired shift assessments and prespecified equivalence criteria will evaluate trait-level preservation and geometric equivalence. A comparison condition, defined uncertainty procedure, and threshold for practical relevance will evaluate the state response. These values motivate the controlled tests in Section 9, with timing, outcome scale, and uncertainty definitions carried into the study design.

5.6 Upstream domain benchmarks: retained separately

The domain atlas provides context around the joint-matrix comparisons. The supplied inventory, labeled ATLAS-001, reports 147 usable entries, with its count and source records assigned to independent audit. The following benchmarks retain their individual observational settings:

- Fluid dynamics: reported critical Reynolds numbers of 2040 ± 10 for puffs and 2550 ± 50 for slugs.
- Cardiac measurements: reported SDNN summaries 129.52 ± 36.12, 141 ± 39, and 50 ± 16; RMSSD 42 ± 15; and DFA exponent $\alpha_1 = 1.231$ versus 0.933 in a comparison involving congestive heart failure. Cohort labels, recording durations, and units must accompany these values before quantitative comparison.
- Neurophysiology: reported alpha peak frequency 10.2 ± 0.9, alpha synchrony $R_\alpha = 0.660$ in the cognitively normal group, and delta synchrony $R_\delta = 0.949$. Reported PLV values are F4–P3 0.44 ± 0.05, C3–C4 0.45 ± 0.02, P8–O1 0.64 ± 0.09, O1–O2 0.6–0.7, and unspecified distant pairs 0.3–0.5. Other reported summaries are entropy 0.66, Lempel–Ziv complexity 0.7 ± 0.1, permutation entropy $H_p = 0.9$ with embedding dimension $m = 3$–5, and Kraskov mutual information 0.25. Bands, estimators, normalizations, and recording conditions require source-specific definitions.
- Bioelectricity: reported membrane potentials -70 versus -10 for a comparison involving proliferative cells; units and cell types must be documented.
- Machine learning: reported Kaplan scaling exponents $N_{\text{exp}} = 0.73$ and $D_{\text{exp}} = 0.27$, compared with Chinchilla values 0.5/0.5; CKA 0.8–0.95; and judge correlation $r_{\text{judge}} = 0.65$. Their datasets, model regimes, and measurement definitions remain distinct.

Each benchmark describes a particular system, measurement, and regime. Cross-domain covariance requires a common observational unit, and critical thresholds and trait–state dimensionality comparisons will be evaluated through their own prespecified tests. The atlas therefore supplies context while the joint samples supply the covariance evidence.

5.7 Claims reserved for prospective validation

The supplied record identifies a further set of claims for prospective validation: cross-dataset cosine similarity 1.00 for the 109-versus-608 comparison; reported AD-versus-CN prediction 0.965 ± 0.043; CHF prediction 0.900 ± 0.094; correlation with a critical Reynolds number $r = 0.903$; and the joint intervention claim $\Delta\mathbf{S} \neq \mathbf{0}$ with approximately preserved traits. Their supporting runs, uncertainty analyses, and replications will establish their evidential status. Dataset identities, prediction metrics, overlap and leakage audits, calibration where applicable, and held-out evaluation form the required validation record. Each claim will proceed through its corresponding prediction, replication, or intervention study.

5.8 Verification requirements

Verification follows the measurement from its source to its interpretation. Each claim will carry its source data, unit hierarchy, feature definitions, preprocessing and scaling, estimator version, and uncertainty procedure. Reconstruction will recover the complete joint matrix and reproduce its spectra and dimensionalities, keeping full-matrix, between-unit, within-unit, bootstrap, and leave-one-unit-out estimates distinct. Longitudinal and intervention claims will additionally carry paired records, timing, comparison conditions, and independently specified reliability or effect criteria.

Discrepancies will become part of the traceable record through comparisons of the relevant data and analyses. Legacy claims will retain provisional status while their supporting records are assembled, and reproduced analyses will establish the basis for their empirical interpretation. The PhysioNet primary test has completed its numerical stage. The next verification stage links its final artifacts, reproduces the run from the frozen record, and conducts feature, projection, and quality-control sensitivity analyses before independent replication.

6 Cognitive Physics Predictions

The following register translates the trait–state framework into claim-specific tests. The first EEG prediction has reached an author-reported result, while the remaining claims retain prospective or provisional status. Future jointly sampled systems may support, reverse, or leave each prediction unresolved. Every participation ratio remains measurement-determined and enters the record at its observed value.

1. **Central dimensionality inequality.**

   The observation model and covariance decomposition are

   $$\begin{aligned} \mathbf{X}_{s,t} &= \boldsymbol{\mu} + \mathbf{T}_s + \mathbf{S}_{s,t} + \boldsymbol{\varepsilon}_{s,t}, \\ \Sigma_{\text{between}} &= \mathrm{Cov}_s(\overline{\mathbf{X}}_s), \\ \Sigma_{\text{within}} &= \mathbb{E}_s\left[\mathrm{Cov}_t(\mathbf{X}_{s,t} - \overline{\mathbf{X}}_s)\right]. \end{aligned}$$

   The primary prediction is

   $$\boxed{d_{\mathrm{PR}}(\Sigma_{\text{between}}) < d_{\mathrm{PR}}(\Sigma_{\text{within}})}. \tag{11}$$

   The completed EEG test supports this ordering with $d_{\text{between}} = 3.54$, $d_{\text{within}} = 6.01$, and $\widehat{\Delta}_d = 2.47$ with bootstrap interval $[1.92, 3.01]$. Earlier reported systems place between-unit dimensionality near 1–1.5 and within-unit dimensionality near 3–10, pending reconstruction of their analytical records. A genuine joint system with $d_{\text{between}} \geq d_{\text{within}}$ supports a reversed ordering and challenges the prediction in that system. The hypothesis fixes the ordering criterion while leaving every dimensionality to measurement.

2. **Completed PhysioNet EEG test.**

   The nominal design allowed approximately 3270×10 joint observations from 109 participant identifiers, 64 channels, 160 Hz recordings, and the eyes-open baseline file pattern represented

by `S001R01.edf`. Quality control retained a 1322×10 matrix from 78 participants, with 31 identifiers excluded after too few accepted epochs remained. The primary inequality was supported: $d_{\text{between}} = 3.54$, $d_{\text{within}} = 6.01$, and $\widehat{\Delta}_d = 2.47$ with bootstrap interval $[1.92, 3.01]$. Leave-one-participant-out analysis remained stable at approximately 2.47 ± 0.04.

Secondary directional predictions concerning PC1 fractions and entropy summaries remain open until their outputs are linked and verified. Raw-band features, feature-family ablations, alternate projections, and quality-control sensitivity analyses will test how strongly the primary result depends on the ten-feature instrument. Every additional estimate will be reported at its measured value:

$$\boxed{d_{\text{PR}}^{\text{full}} \text{ is determined by the retained data.}} \tag{12}$$

3. **Longitudinal stability.**

   Candidate trait summaries are predicted to remain strongly ordered across visits, with $\rho(t_0, t_1) > 0.85$ and $\rho(t_0, t_2) > 0.8$ across months. The reported precedents are 0.969, 0.919, and 0.876 across the three visit pairs. Candidate state summaries are predicted to show weaker longitudinal correlation, with $\rho < 0.2$; the reported state value is 0.018. Reliability intervals and visit-specific quality controls will determine the strength of this evidence.

4. **Predictive validity.**

   Trait-space summaries are predicted to carry held-out information about prespecified outcomes. The supplied targets include AD-versus-CN cross-validated performance in the range 0.90–0.97, with a previous summary of 0.965 ± 0.043 and an associated alignment near $r \sim 0.99$; CHF performance of 0.900 ± 0.094; and association with critical Reynolds number of $r = 0.903$. These claims require recoverable runs, participant-level validation, uncertainty analysis, calibration where applicable, and leakage audits before empirical acceptance.

5. **Selective intervention response.**

   For a perturbation directed at transient state, the prediction is

$$\boxed{\Delta \mathbf{S} \neq \mathbf{0}, \qquad \mathbf{T}_{\text{pre}} \approx \mathbf{T}_{\text{post}}}. \tag{13}$$

   The reported tDCS precedent includes $N = 30$, trait correlation $\rho(\text{pre}, \text{post}) = 0.998$, mean state change 0.124 ± 0.212, state entropy 2.30, and metastability 0.154 ± 0.057. For an effective AD intervention, a further prediction is movement of the prespecified trait summary toward the cognitively normal reference $R_\alpha = 0.660$. A trait change exceeding the prespecified equivalence margin challenges the selective-response claim for that intervention. Prospective replication will determine its scope.

6. **Cross-trait coupling on a genuine common unit.**

   For the 58-patient neuro-cardiac matrix, a shared systemic factor predicts weak positive coupling, $0.2 \leq \rho(\text{trait}_{\text{EEG}}, \text{trait}_{\text{cardiac}}) \leq 0.4$; the reported value is 0.297. A result near zero supports relative independence of the measured traits. As a row-semantics diagnostic, independently sampled measurement families are predicted to lose their apparent compression when their synthetic alignment is broken: the supplied comparison moves from $d_{\text{PR}} = 2.28$ to 4.19–4.26. This prediction expresses the principle that covariance rests on observational coexistence.

7. **System-specific spectra and cross-dataset generalization.**

   Spectral decay is predicted to retain system-specific form. The reported exponents are $\alpha = 3.92$ for EEG, 3.61 for cardiac data, and 4.90 for ML. Under a shared feature pipeline, trait-loading directions across independent datasets are predicted to have cosine similarity above 0.9; the supplied 109-versus-608 comparison reports 1.00. Pipeline changes are expected to alter that alignment, linking geometric generalization to the representation used to construct it.

8. **Structure-destroying null tests.**

   When a persistent trait organizes a genuine joint matrix, the observed geometry is predicted to change under a shuffle that disrupts the unit-to-observation relationship while preserving specified marginal structure. In the completed EEG run, participant-label shuffling reduced the mean dimensionality contrast from 2.47 to approximately 0.83, indicating that the participant hierarchy contributes to the full effect while also revealing residual structure that requires further null calibration. Random one-dimensional projections yielded contrasts near zero, locating the primary separation in the multivariate feature geometry. Earlier full-matrix precedents report EEG $1.48 \ll 9.949 \pm 0.011$, cardiac $1.64 \ll 6.663 \pm 0.090$, and ML $1.33 \ll 4.767 \pm 0.090$, pending reconstruction. Each analysis will define the disrupted relationship and preserved structure of its null. A precisely estimated absence of the predicted separation challenges the corresponding persistent-compression interpretation.

Together, these predictions turn the framework into a continuing measurement program. The first prospective EEG test supports the central ordering in its declared feature space. Artifact linkage, robustness analyses, independent replication, longitudinal reliability, prediction, and intervention now provide the next encounters with evidence. Each claim survives in proportion to the support supplied by newly sampled joint data.

7 EEG Study Protocol and Execution

The EEG study carried the geometric question into a defined recording condition. The protocol was established before inspection of the primary covariance spectra or dimensionality difference. Its reported core used common-average referencing, 1–30 Hz FIR filtering, nonoverlapping two-second epochs, and alpha-phase extraction through an 8–13 Hz fourth-order Butterworth filter followed by the Hilbert transform. The completed run evaluated the primary ordering within this frame. Archiving the frozen implementation, version record, joint feature matrix, exclusion ledger, and final machine-readable output will complete the link between protocol and result.

7.1 Dataset and observational hierarchy

The resource was the PhysioNet EEG Motor Movement/Imagery Dataset, version 1.0.0, with EDF+ recordings from 64 EEG channels sampled at 160 Hz [1]. Its directory contains 109 participant identifiers, and the baseline protocol includes one-minute eyes-open and eyes-closed recordings. The primary study selected the eyes-open baseline run, `R01`; `S001R01.edf` illustrates the file naming

convention. Recording availability and implemented quality checks reduced the initial inventory to 78 retained participants.

Participants defined the units s, and nonoverlapping two-second epochs defined the repeated observations t. Each vector gathered features from a single participant and accepted epoch, preserving the shared window that gives joint measurement its meaning. A nominal one-minute recording allowed approximately 30 epochs per participant, or approximately $109 \times 30 = 3270$ candidate rows. Quality control retained 1,322 epochs. Thirty-one participant identifiers were excluded after too few accepted epochs remained for the declared covariance comparison.

7.2 Preprocessing and quality control

The reported preprocessing sequence comprised channel-label harmonization, channel and recording quality checks, common-average referencing over retained EEG channels, and 1–30 Hz finite-impulse-response filtering before epoching. Alpha-phase features used an additional 8–13 Hz fourth-order Butterworth filter followed by the Hilbert transform. The archived implementation will verify the reference channel set, filter design, forward or bidirectional application, padding, transient treatment, and channel-handling rules. These details preserve a traceable route from each continuous recording to each accepted feature vector.

Implemented artifact thresholds, usable-duration criteria, minimum retained epochs, and channel rules governed retention before covariance estimation. Their combined effect yielded the 78-participant, 1,322-epoch matrix. The exclusion ledger will preserve each decision and connect the 31 excluded identifiers to explicit criteria. Secondary analyses with documented alternative thresholds will examine how participant and epoch retention shape the dimensionality estimates.

7.3 Joint feature vector

The reported ten-feature primary vector contains:

1. Four band-power summaries, averaged across the selected channel set: 1–4 Hz, 4–8 Hz, 8–13 Hz, and 13–30 Hz. Archive review will verify the spectral estimator, treatment of band edges, and power transformation.

2. Two alpha-phase summaries: the epoch mean and temporal standard deviation of the Kuramoto order parameter,
$$R(u) = \left| \frac{1}{K} \sum_{k=1}^{K} e^{\mathrm{i}\theta_k(u)} \right|, \tag{14}$$
where K is the number of included channels and u indexes samples within an epoch.

3. One alpha phase-locking value for the prespecified F4–P3 channel pair,
$$\mathrm{PLV}_{\mathrm{F4,P3}} = \left| \frac{1}{M} \sum_{u=1}^{M} e^{\mathrm{i}[\theta_{\mathrm{F4}}(u) - \theta_{\mathrm{P3}}(u)]} \right|, \tag{15}$$

where M is the number of valid samples in the epoch.

4. One normalized eigenvalue entropy of the within-epoch channel covariance, defined for a nonzero covariance by

$$H_{\mathrm{eig}} = -\frac{1}{\log K} \sum_{k=1}^{K} q_k \log q_k, \qquad q_k = \frac{\lambda_k}{\sum_j \lambda_j}, \tag{16}$$

with $0 \log 0 = 0$. This is a channel-level epoch feature, distinct from the between-unit and within-unit feature covariances.

5. Two complexity summaries: channel-averaged Lempel–Ziv complexity and normalized permutation entropy. Archive review will verify the signal inputs, symbolization, normalization, embedding, delay, and tie handling.

Each estimator retains a clear measurement identity. Eigenvalue entropy and permutation entropy carry distinct names and remain linked to the signal or covariance being summarized. The next implementation audit will recover synthetic-signal checks, short-epoch behavior, and finite-sample bias assessments. The released joint matrix will carry participant identifiers, epoch boundaries, and quality-control status, preserving the connection between numerical features and their observational origin.

7.4 Primary analysis and uncertainty

The accepted feature matrix yielded the two covariance geometries through Equations (5) and (6). A common scaling rule and equal participant weighting placed them in a shared coordinate frame. The run produced $d_{\mathrm{PR}} = 3.54$ between participants, $d_{\mathrm{PR}} = 6.01$ within participants, and $\widehat{\Delta}_d = 2.47$. The archived code and matrix will verify the participant-balanced scaling implementation and treatment of constant or undefined features. Covariance spectra and first-component variance fractions remain outputs for archival extraction and reporting.

The primary uncertainty procedure used a participant-level bootstrap with 2000 resamples, as recorded in the supplied execution log. The reported 95% interval for Δ_d was $[1.92, 3.01]$. Leave-one-participant-out analysis yielded approximately 2.47 ± 0.04. The archive audit will verify that each bootstrap draw preserved complete accepted epoch sequences and recomputed every data-dependent transformation according to protocol. Independent replication will extend the comparison to separate data, while additional temporal resampling and surrogate analyses will carry explicitly specified dependence models and null hypotheses.

7.5 Decision rules and scope

The prespecified decision rule assigned support when the full uncertainty interval lay above zero, a reversed ordering when the full interval lay below zero, and an unresolved result when the interval crossed zero. The observed interval $[1.92, 3.01]$ therefore supported the primary prediction. Interpretation as latent traits or states continues to require a measurement-error account and

independent reliability evidence. Split-window analyses can examine short-timescale stability within recordings, while independent-session data will evaluate persistence across visits. Eyes-closed recordings retain a separate condition label and separately specified covariance analysis.

The released analysis record will carry the path from source recording to geometric comparison: the frozen protocol, data version and file manifest, software versions, random seeds, exclusion ledger, joint feature matrix, and uncertainty outputs, subject to the dataset's terms. The primary R01 analysis has been completed, while archival verification and additional sensitivity analyses remain in progress. Independent-session replication, clinical prediction, and intervention response will proceed through subsequent studies, each with its own design and evidential outcome.

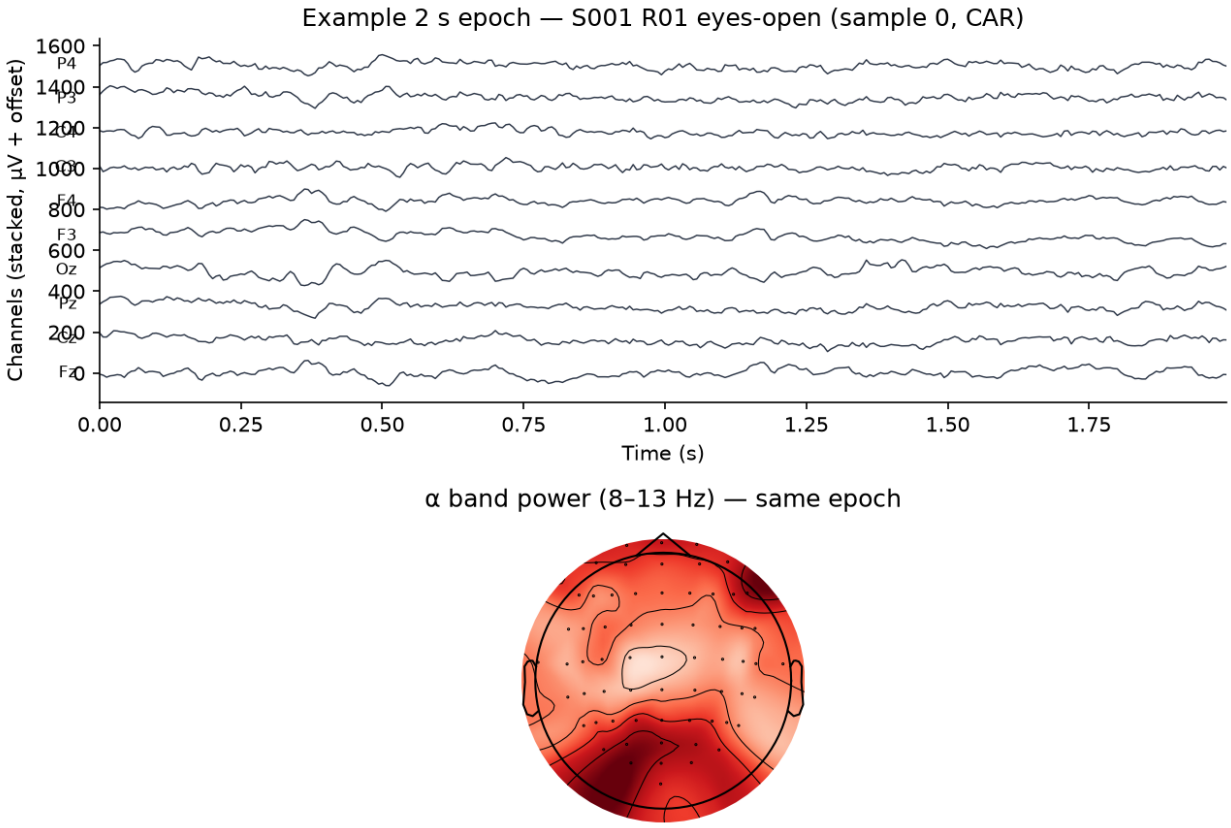


Figure 1: Example measurement-level EEG visualization from participant S001, eyes-open run R01. The upper panel shows a common-average-referenced two-second epoch for selected channels, displayed with vertical offsets. The lower panel shows 8–13 Hz band power over the scalp for the same epoch. This figure illustrates the source signal and spatial structure from which epoch-level features are derived; the primary covariance comparison uses the declared ten-feature vectors.

8 Primary EEG Results

This section records the numerical outcome supplied from the completed PhysioNet R01 run. The values are treated as author-reported until the final `RESULT_LATEST.json`, frozen code version,

exclusion ledger, and joint feature matrix are linked within the manuscript archive. The supplied execution log records 2000 bootstrap resamples and 1000 null permutations. This provenance status concerns verification of the computational record; the decision rule and interpretation below follow the prespecified comparison.

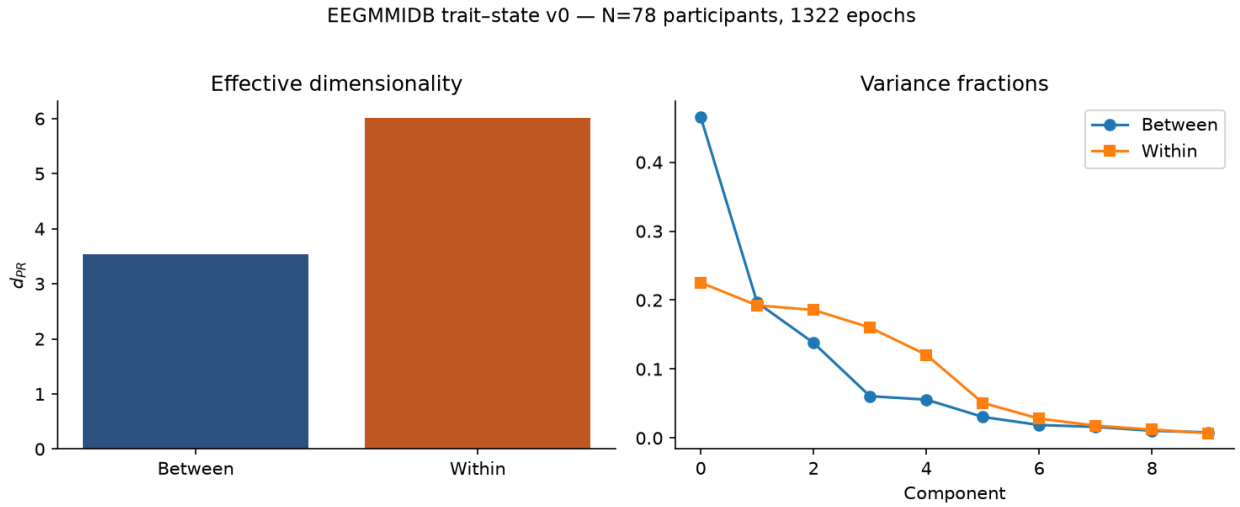


Figure 2: Summary of the completed EEGMMIDB R01 covariance comparison. The left panel shows participation-ratio effective dimensionality for between-participant and within-participant covariance. The right panel shows the normalized variance fractions across the ten covariance components, revealing a more concentrated leading direction between participants and a more distributed within-participant spectrum.

8.1 Retained sample and primary comparison

Quality control retained 78 participants and 1,322 two-second epochs in the ten-feature joint matrix. Thirty-one participant identifiers contributed too few accepted epochs under the implemented criteria and were excluded from the covariance comparison. The primary estimates were

$$\boxed{d_{\mathrm{PR}}(\widehat{\Sigma}_{\mathrm{between}}) = 3.54, \quad d_{\mathrm{PR}}(\widehat{\Sigma}_{\mathrm{within}}) = 6.01, \quad \widehat{\Delta}_d = 2.47.} \tag{17}$$

The participant-level bootstrap produced a 95% interval of $[1.92, 3.01]$. Because the full interval lies above zero, the run satisfies the preregistered criterion for supporting the primary ordering,

$$d_{\mathrm{PR}}(\widehat{\Sigma}_{\mathrm{between}}) < d_{\mathrm{PR}}(\widehat{\Sigma}_{\mathrm{within}}). \tag{18}$$

The registered verdict is `supports_primary`. This verdict describes the geometry measured in the retained cohort under the implemented quality-control rules, scaling procedure, and ten-feature representation.

8.2 Stress checks

Leave-one-participant-out analysis yielded a mean dimensionality difference of approximately 2.47 ± 0.04, showing that the observed ordering remained stable as individual participants were removed in

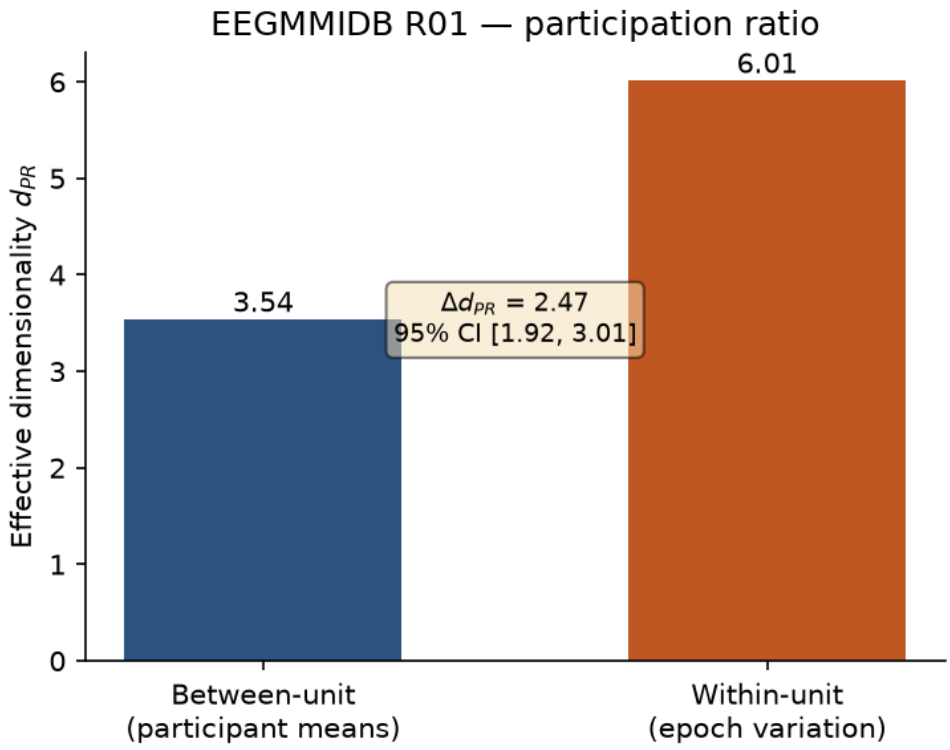


Figure 3: Primary effective-dimensionality comparison. Within-participant epoch variation yielded $d_{\mathrm{PR}} = 6.01$, while covariance across participant means yielded $d_{\mathrm{PR}} = 3.54$. Their difference was $\widehat{\Delta}_d = 2.47$, with participant-bootstrap 95% interval $[1.92, 3.01]$.

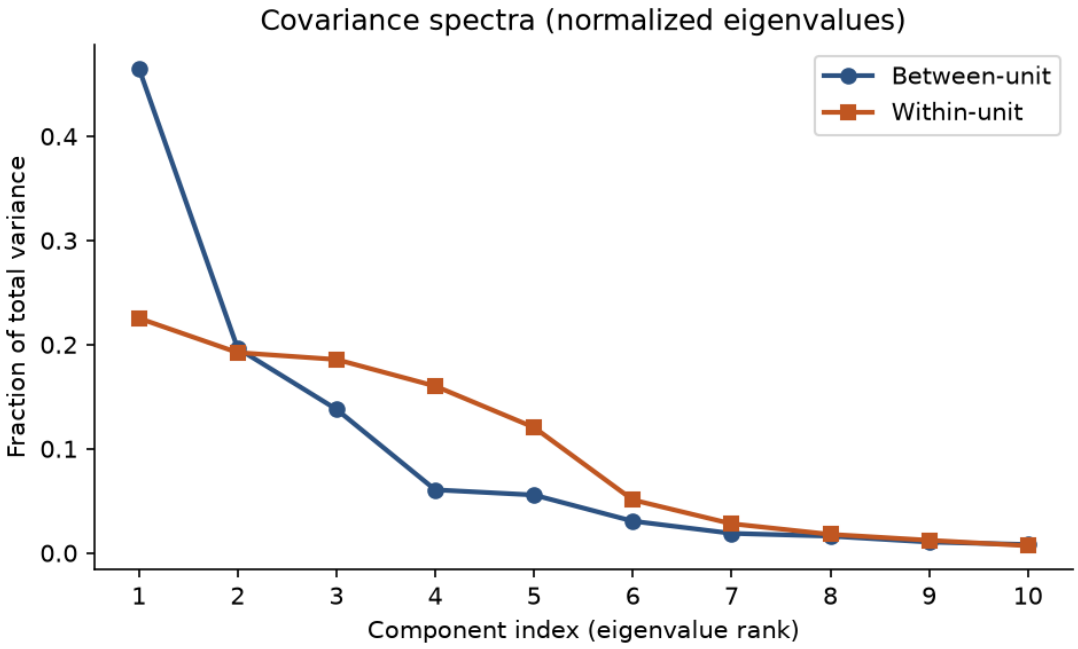


Figure 4: Normalized covariance spectra for the ten-feature representation. Between-participant variance is concentrated in its leading component. Within-participant variance spreads across more components, producing the larger participation ratio.

turn. A participant-label shuffle reduced the mean difference to approximately 0.83. The reduction indicates that the participant-to-epoch hierarchy contributes materially to the full separation. The positive shuffled mean also shows that the chosen marginals, scaling, or feature geometry can generate part of the contrast after identity is disrupted. A complete null report will therefore include the permutation distribution, interval or tail probability, preserved structure, and exact relabeling procedure.

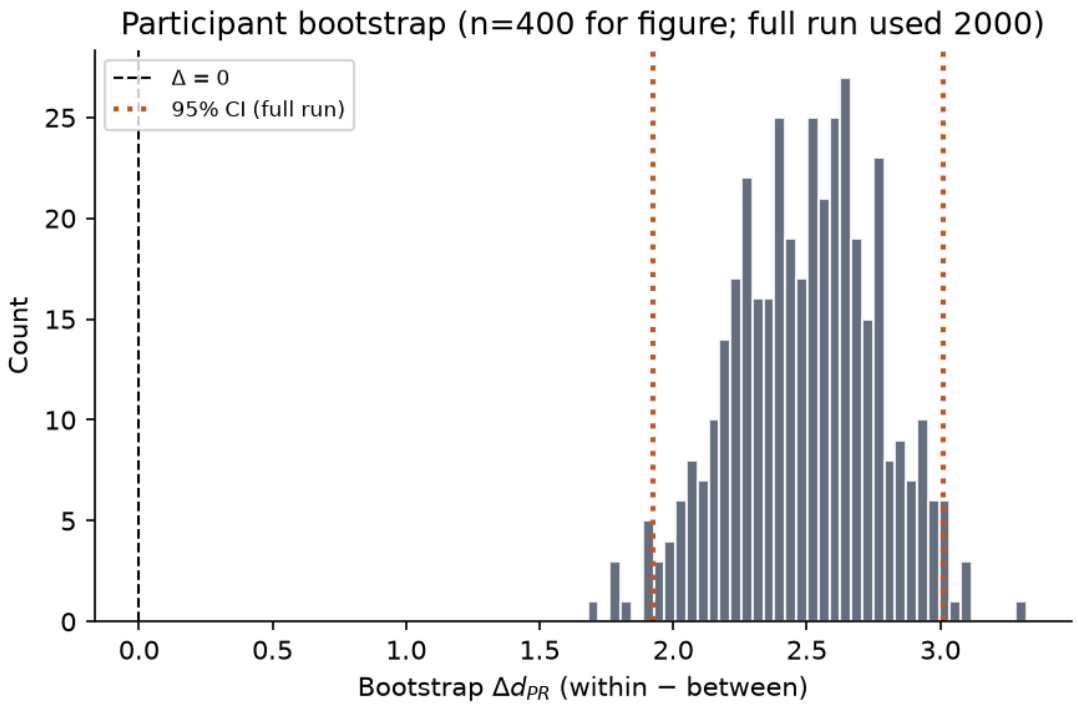


Figure 5: Participant-bootstrap distribution of $\Delta d_{\mathrm{PR}} = d_{\mathrm{PR,within}} - d_{\mathrm{PR,between}}$. The displayed histogram uses 400 resamples for visualization, while the dotted interval markers show the 95% interval $[1.92, 3.01]$ from the full 2000-resample run. The zero reference lies well below the reported interval.

Random one-dimensional projections produced dimensionality differences near zero, as expected when each covariance is represented by a single variance direction. The observed contrast therefore belongs to the multivariate geometry of the ten-feature design. This dependence forms part of the measurement claim: the selected features act as an instrument through which between-participant and within-participant structures become visible.

8.3 Interpretive scope and next checks

The completed run supports the predicted dimensionality ordering for eyes-open R01 recordings in the analyzed feature space. The narrow bootstrap interval and stable leave-one-participant-out estimates support precision and within-sample robustness. The shuffle and projection analyses locate important sources of the effect in participant identity and multivariate representation. Trait and state interpretations continue to depend on reliability across sessions, measurement-error analysis, and replication. The 31 quality-control exclusions also motivate sensitivity analysis with documented alternative thresholds to determine how cohort retention shapes the estimates.

The next robustness analyses will compare raw or minimally transformed band-power features, feature-family ablations, alternative projections, and quality-control settings. A separately specified condition or dataset will provide the stronger replication test. Together, these analyses will

distinguish geometry that travels across reasonable measurement choices from geometry that is specific to the present instrument.

9 Prediction, Intervention, and Falsification Tests

Persistence asks what endures across observations; prediction asks what information transfers to new cases; intervention asks what responds to a controlled change. The R01 result provides a geometric starting point: within-participant covariance occupied more effective dimensions than between-participant covariance in the declared ten-feature space. Robustness, persistence, utility, and response remain distinct claims. Each extension will proceed through a dedicated study with independently specified outcomes and decision criteria, allowing interpretation to develop alongside evidence.

9.1 Robustness before extension

The first task is to establish how the supported ordering responds to reasonable changes in the measurement instrument. A clean rerun from the frozen archive will test computational reproduction. Prespecified sensitivity analyses will then examine raw or minimally transformed band-power features, removal of phase, entropy, or complexity families, alternative participant-balanced scaling, documented quality-control thresholds, and projections of varying dimension. Each analysis will retain the same participant hierarchy and report both dimensionalities, their difference, and uncertainty.

An ordering that remains positive across these analyses will support robustness over the tested measurement choices. A contrast that narrows, becomes unresolved, or reverses will identify the representation or retention rules that carry the result. Such an outcome refines the claim by locating its domain. The participant-label null will also receive a full permutation distribution and explicit statement of preserved marginals, allowing the observed 2.47 contrast and shuffled mean near 0.83 to be compared on a calibrated scale.

9.2 Persistence across independent sessions

The completed R01 analysis measures variation within a single recording condition. Repeated recordings will follow the same units across independent sessions under matched reference conditions. Common feature extraction and scaling will preserve the coordinate frame of the comparison. Prespecified outcomes will characterize the reliability of unit-level summaries and the reproducibility of between-unit covariance spectra and directions. Persistence will be evaluated through this combined record: effective dimensionality, principal directions, and individual unit positions each describe a distinct aspect of the geometry.

The study will distinguish the continuity of population geometry from the stability of individual differences. Acquisition conditions, recording duration, and prespecified nuisance variables will

accompany the measurements. Adjacent windows will describe repeatability within a recording, while independent visits will extend the assessment across longer intervals. The timescale of each observation will define the timescale of the persistence claim.

9.3 Held-out predictive value

Prediction gives the measured geometry a question beyond the sample in which it was constructed. The study will prespecify its target, prediction horizon, baseline comparators, performance measures, and minimum useful improvement. Trait-like and state-like summaries will be evaluated separately and jointly to assess their additional information. Feature scaling, geometric projections, feature selection, and hyperparameter tuning will be fitted within training data; entire held-out units will supply the evaluation sample. For time-dependent targets, earlier predictors and later outcomes will retain their temporal separation. Earlier AD, CHF, and cross-domain prediction summaries remain provisional precedents until their runs and validation records are reconstructed.

The report will present uncertainty and, where appropriate, calibration alongside discrimination or prediction error. An independent cohort will evaluate transportability. Predictive utility will be judged against the prespecified improvement over baseline and its uncertainty, with the held-out observations determining performance. Geometric reproducibility and predictive value will retain separate evidential records, connecting the shape of variation to its practical reach through explicit tests.

9.4 Controlled intervention test

A controlled perturbation brings the geometry into a setting where response can be measured. The earlier tDCS summary—trait correlation 0.998 and mean state change 0.124 ± 0.212 in 30 participants—serves as a provisional precedent whose source record remains to be reconstructed. For a future intervention hypothesized in advance to produce a transient state response, the proposed prediction is

$$\Delta \mathbf{S} \neq \mathbf{0}, \qquad \mathbf{T}_{\text{post}} \approx \mathbf{T}_{\text{pre}}. \tag{19}$$

The hypothesis applies to prespecified perturbations expected to act on transient state; interventions intended to produce durable changes will receive separately formulated hypotheses. Repeated baseline recordings will establish the reference condition, and prespecified post-intervention windows and follow-up sessions will characterize the response over time. A randomized comparison condition will be used where feasible. Assignment, timing, and analysis will be specified before inspection of the outcomes.

Identification of the latent components in Equation (3) will rest on explicit assumptions and repeated-measurement evidence. The operational state outcome will therefore be defined independently of the treatment result. Let $\mathbf{Z}_{s,\text{pre}}$ and $\mathbf{Z}_{s,\text{post}}$ denote prespecified state-sensitive summaries obtained with a common feature map fitted on independent baseline or training data. For assignment $A_s \in \{0, 1\}$, the operational intervention contrast is

$$\begin{aligned} \boldsymbol{\tau}_S = \mathbb{E}[\mathbf{Z}_{s,\text{post}} - \mathbf{Z}_{s,\text{pre}} \mid A_s = 1] \\ - \mathbb{E}[\mathbf{Z}_{s,\text{post}} - \mathbf{Z}_{s,\text{pre}} \mid A_s = 0]. \end{aligned} \tag{20}$$

A primary scalar contrast or multivariate decision rule, minimum relevant effect, and unit-respecting uncertainty procedure will be preregistered. The feature map will preserve the condition-level mean shifts targeted by this contrast. Interpreting a controlled change in $\mathbf{Z}$ as a latent state response will additionally require identification assumptions, repeated-session evidence, and follow-up observations that distinguish trait, state, and residual contributions. The operational response and its latent interpretation will remain linked through these explicit evidential steps.

9.5 Establish approximate invariance through equivalence criteria

Preservation is itself a measurable claim. Approximate invariance of trait-related geometry will be evaluated through a prespecified equivalence criterion with a precision requirement appropriate to the comparison. One proposed discrepancy, computed under common scaling, is

$$g_T = \frac{\|\Sigma_{\text{between,post}} - \Sigma_{\text{between,pre}}\|_{\text{F}}}{\|\Sigma_{\text{between,pre}}\|_{\text{F}}}, \tag{21}$$

where $\|\cdot\|_{\text{F}}$ is the Frobenius norm and the baseline covariance must be nonzero. A tolerance δ_T will be fixed from the scientific question and independent repeatability information. Geometric equivalence will require a prespecified upper confidence bound for g_T below δ_T. Paired unit-level reliability and shift assessments will accompany this population-level comparison. Covariance preservation, stability of individual positions, and preservation of the population mean will each receive a distinct assessment.

The selective-response hypothesis requires state sensitivity and trait preservation to meet their respective criteria together. An imprecise comparison will remain unresolved. A consequential trait-related change or a precisely estimated state response below the required threshold will challenge the corresponding claim. The interpretation will follow the prespecified hypothesis and the precision with which its criteria can be evaluated.

9.6 Claim-specific falsification

Each test gives the framework a defined encounter with evidence. The report will identify the outcome at the level of the claim examined:

1. The present R01 run supports the primary ordering in the declared ten-feature space; exact reproduction from the archived artifacts will establish computational traceability.
2. An unresolved or reversed ordering under raw-band, feature-ablation, scaling, or quality-control analyses will constrain the robustness claim to the representations that retain support.
3. A reliably negative Δ_d in a separately specified replication supports a reversed dimensionality ordering and challenges broader extension of the central prediction.
4. Independent-session reliability below the required criterion challenges the persistent-trait interpretation over the tested timescale.

5. Held-out improvement below the prespecified utility threshold challenges the predictive application; the geometric comparison retains its own evidential assessment.

6. Precise evidence inconsistent with the joint state-response and trait-preservation criteria challenges the restricted intervention hypothesis; an imprecise result remains unresolved.

The sampled populations, feature definitions, conditions, and timescales will define the scope of each outcome. Additional tests will establish the evidence for broader extensions. The program will advance through reproducible measurements and claim-specific evaluation, with each interpretation revised in proportion to what the observations support.

10 Limitations, Contributions, and Next Tests

10.1 Limits of the decomposition and measurement design

Every measurement has a horizon defined by its sampling, representation, and precision. The additive trait–state model provides a working frame within that horizon, with identification grounded in explicit assumptions and supporting observations. Persistent context, anatomy, acquisition differences, slow state variation, and measurement error may shape the estimated between-participant geometry. Within-participant geometry may combine state fluctuations with residual noise, artifacts, and short-timescale dependence. The measured covariance matrices therefore establish an operational geometric distinction; independent-session reliability and controlled perturbations will connect that distinction to persistence and response.

The primary EEG result comes from one eyes-open baseline condition and approximately one minute of recording per available participant. Quality control retained 78 of 109 participant identifiers and 1,322 epochs. The 31 exclusions may alter the sampled population if recording quality covaries with participant characteristics or physiology. Alternative quality-control thresholds, retention summaries, and comparisons between included and excluded records will characterize this source of selection. Temporal dependence among adjacent epochs also shapes the effective information in the within-participant estimate and motivates block or surrogate analyses alongside participant-level resampling.

The bootstrap interval $[1.92, 3.01]$ quantifies the declared participant-resampling procedure and supports the ordering within the retained sample. The participant-label shuffle retained a positive mean contrast near 0.83, indicating that feature marginals, scaling, or residual geometry contribute to the statistic after identity is disrupted. A calibrated permutation distribution and explicit null construction are therefore central to interpretation. The final numerical record also retains author-reported status while the result JSON, frozen code, matrix, exclusion ledger, and environment are linked and rerun.

10.2 Feature dependence and limits of generalization

The measured geometry takes its form within a chosen feature space. Effective dimensionality therefore retains the context of its features, scaling, preprocessing, and sampling timescale. The random one-dimensional projection result, with Δ_d near zero, shows that the observed separation lives in multivariate structure. Several summaries drawn from the same EEG may also emphasize overlapping information. Raw-band analyses, feature-family ablations, alternative projections, and scaling comparisons will map how the result changes as the measurement instrument changes.

Extension across systems requires explicit correspondence between measured quantities, replication within each system, and evidence addressing the proposed mechanism. Predictive performance will characterize statistical utility, while an appropriate causal design will support causal interpretation. Clinical discrimination, treatment efficacy, and individual decision support will each proceed through dedicated validation beyond the initial resting-EEG comparison. The scope of a conclusion will grow with the evidence supporting its extension.

Earlier EEG, cardiac, machine-learning, longitudinal, predictive, and intervention summaries retain provisional status while their supporting data and analysis records are recovered. The completed R01 result carries stronger procedural standing through its prospective decision rule, while full computational verification awaits archive linkage. Cognitive Physics names the measurement program developed here, with a present focus on the geometry of jointly observed features. The present evidence supports a bounded feature-space claim. Claims concerning physical laws, cognition, or general trait–state separation require separately formulated hypotheses and evidence suited to those questions.

10.3 Contributions and remaining deliverables

The study has begun a traceable passage from observations to covariance geometry and from geometry to tested interpretation. Its present and remaining outputs are:

1. A completed prospective EEG comparison with a declared observational hierarchy, primary inequality, participant-level bootstrap, and decision rule.
2. An author-reported result of $d_{\mathrm{PR}} = 3.54$ between participants and 6.01 within participants, with $\widehat{\Delta}_d = 2.47$ and 95% interval $[1.92, 3.01]$.
3. Stress checks showing stable leave-one-participant-out estimates, a reduced contrast under participant-label shuffling, and collapse toward zero under one-dimensional projection.
4. A remaining archival package containing the frozen code, final JSON, software environment, input manifest, exclusion ledger, joint feature matrix, spectra, and uncertainty outputs.
5. A robustness and replication program covering feature ablations, quality-control sensitivity, calibrated null tests, independent sessions, held-out prediction, and controlled intervention.

The first test supports the central ordering in its sampled system and feature space. A reversed or unresolved result in a future system will constrain the scope of that support and identify where

the geometry changes. Each outcome contributes a clear record of what was measured, how its interpretation was evaluated, and which observations would advance the inquiry. Methodological novelty and broader scientific significance will be assessed through comparison with relevant existing work and evidence of the contribution achieved. The program's enduring output is intended to be a reproducible relationship between measurements, methods, and claims.

11 Conclusion

The geometry of persistence and the movement of state gave this study a direct question: within a jointly observed system, does variation across unit means occupy fewer effective dimensions than variation around those means? The first prospective EEG test answered that question for one declared measurement setting. In 78 participants and 1,322 accepted epochs, the author-reported analysis yielded $d_{\mathrm{PR}} = 3.54$ between participants and 6.01 within participants, with $\widehat{\Delta}_d = 2.47$ and a bootstrap 95% interval of $[1.92, 3.01]$. Under the prespecified decision rule, the result supports the central ordering in the ten-feature eyes-open R01 analysis.

The surrounding checks give the result its proper shape. Leave-one-participant-out estimates remained stable near 2.47 ± 0.04. Participant-label shuffling reduced the mean contrast to approximately 0.83, revealing contributions from participant identity alongside residual feature or marginal structure. One-dimensional projections moved the contrast toward zero, locating the separation in multivariate geometry. The finding therefore belongs to the dataset, quality-control rules, scaling, and feature representation through which it was measured. Its present standing is a bounded covariance result; persistent-trait, predictive, causal, and physical-law interpretations require their own evidence.

The method combines three established ingredients: multilevel covariance decomposition, participation ratio, and unit-respecting uncertainty. The resulting statistic, Δd_{PR}, converts the relative concentration of two covariance spectra into a directional repeated-measures hypothesis. Evidence can support the ordering, support its reversal, or leave it unresolved. The contribution lies in this synthesis and testing framework rather than in the separate invention of trait–state decomposition or participation ratio.

The next passage runs from reported result to durable record. Linking the frozen code, final JSON, environment, input manifest, exclusion ledger, joint matrix, spectra, and uncertainty outputs will establish computational traceability. Feature ablations, calibrated null tests, alternative quality-control rules, and separately specified replication will test robustness. Independent sessions, held-out prediction, and controlled intervention will then examine persistence, utility, and selective response. Cognitive Physics advances through this sequence: measure each system in its own setting, compare clearly defined geometries, invite outcomes that support or reverse the prediction, and let the reach of every claim follow the reach of its evidence.

Tools Used and AI Assistance

The author used Meta AI (Meta) and ChatGPT (OpenAI) during manuscript development for brainstorming, drafting support, organization, and language revision. The computational EEG experiment was conducted and completed inside Cursor with Grok 4.7 and Composer 2.5 assisting code development, execution, debugging, run monitoring, and summarization of the generated outputs. The OpenAI Codex assistant operating in the Prism manuscript workspace contributed direct LaTeX editing, mathematical and stylistic revision, integration of author-supplied run summaries, document organization, formatting checks, and repeated compilation of the manuscript. These tools assisted the research and preparation process and form part of the manuscript's provenance record.

Empirical claims derive from the identified datasets, author-supplied execution records, and associated analysis artifacts. AI assistance served writing, editing, and software-support roles; computational verification remains assigned to the frozen code, final result files, joint matrices, exclusion records, and reproducible reruns. The author reviewed the manuscript and retains responsibility for the study design, scientific claims, numerical accuracy, source attribution, interpretation, reproducibility requirements, and final text. Tool names, access period, and relevant interaction records can accompany a journal-specific disclosure or archival supplement.

References

[1] G. Schalk. EEG Motor Movement/Imagery Dataset, version 1.0.0. PhysioNet, 2009. DOI: 10.13026/C28G6P.

[2] C.-Z. Di, C. M. Crainiceanu, B. S. Caffo, and N. M. Punjabi. Multilevel functional principal component analysis. *The Annals of Applied Statistics*, 3(1):458–488, 2009. DOI: 10.1214/08-AOAS206SUPP.

[3] E. Campos et al. Multilevel hybrid principal components analysis for region-referenced functional EEG data. *Statistics in Medicine*, 41(19):3737–3757, 2022. DOI: 10.1002/sim.9445.

[4] E. L. Hamaker, N. K. Schuurman, and E. Zijlmans. Using a few snapshots to distinguish mountains from waves: Weak factorial invariance in the context of trait–state research. *Multivariate Behavioral Research*, 52(1):47–60, 2017. DOI: 10.1080/00273171.2016.1251299.

[5] P. Gao, E. Trautmann, B. Yu, G. Santhanam, S. Ryu, K. Shenoy, and S. Ganguli. A theory of multineuronal dimensionality, dynamics and measurement. *bioRxiv*, 2017. DOI: 10.1101/214262.

[6] Y. Hu and H. Sompolinsky. The spectrum of covariance matrices of randomly connected recurrent neuronal networks with linear dynamics. *PLoS Computational Biology*, 18(7):e1010327, 2022. DOI: 10.1371/journal.pcbi.1010327.